

Sea ice, extremophiles and life on extra-terrestrial ocean worlds

<https://www.cambridge.org/core/journals/international-journal-of-astrobiology/article/sea-ice-extremophiles-and-life-on-extraterrestrial-ocean-worlds/C76FF80A75B755492331A3356CD1B824>

The review by Andrew Martin and Andrew McMinn takes us on an exciting journey into the world of sea-ice microbes and their potential role in the search for life on ocean worlds like Europa and Enceladus. These intriguing moons are on the radar for astrobiologists because they're thought to have hidden oceans that have been around for billions of years. Although the conditions there might seem harsh and unfriendly to life, there could still be some warm, welcoming corners associated with the ice, the ocean beneath, and the ocean floor.

One of the big challenges is figuring out where these potential energy sources could come from, as well as understanding how life might metabolize and create ecosystems in such extreme conditions. The authors highlight how adaptable sea-ice microorganisms are, showing just how resilient life can be in the face of tough challenges. These tiny creatures thrive in extreme environments here on Earth, making them excellent stand-ins for what life could be like on other planets.

Microorganisms are incredibly important for our planet's health, and they play a key role in energy transfer and important chemical cycles in our oceans. Believe it or not, they make up a biomass that's several times larger than all the multicellular marine life combined! With billions of bacteria estimated to exist, they outnumber stars in the universe by a long shot. Given that microbes appeared on Earth about 3 billion years before land plants, their diversity and ability to adapt are truly fascinating. Even though we've learned a lot, there's still so much we don't know about them, like how they can provide valuable genes for medical and biotechnological advancements.

The authors also point out that if life can arise through natural processes, then microbes might be a universal foundation for life in the universe. This means that if there's life on other celestial bodies, it could resemble what we see here on Earth. However, if the conditions on moons like Europa or even Mars don't allow for any life at all, then the insights from our microbial friends might not apply.

Understanding what makes life possible in Earth's most extreme environments has really sparked interest in the field of astrobiology. While we're still figuring out how and when life began on Earth, studying organisms that thrive in challenging conditions can offer clues about the potential for life elsewhere in the universe.

The authors adopt a very straightforward definition of habitability proposed by Cockell et al. (2016), which suggests that a habitat must support at least one type of known organism. This practical approach keeps things simple and avoids the tricky business of trying to define what "life" really is. Additionally, this concept connects closely to how far habitability extends, giving us context about the organisms that might be present and how long they can thrive.

In short, the insights we gather from Earth's extremophiles can be incredibly valuable as we seek to understand the possibilities for life in the mysterious environments beyond our planet.

How does salinity shape ocean circulation and ice geometry on Enceladus and other icy satellites?

<https://www.science.org/doi/10.1126/sciadv.abm4665>

The recent research by Wanying Kang and her team closely examines how salinity affects ocean circulation and ice structure on Enceladus, one of Saturn's fascinating moons. This moon is fascinating for scientists, as it features a global saline ocean underneath a thick ice shell, hinting at interesting interactions between water and rock that could make it a candidate for habitability.

****Here are some of the key highlights from the study: ****

- **Ocean Dynamics and Salinity**:** The researchers wanted to understand how the saltiness of the ocean and the way heat is distributed between the moon's rocky core and its icy surface impact ocean movement and temperature regulation. They discovered that the salinity greatly influences how the ocean circulates. Interestingly, at very low and very high salinity levels, the circulation patterns behave quite differently, showcasing just how sensitive the ocean is to changes in salt content. In both cases, fresh water and heat tend to flow toward the equator, where the ice is thicker, helping to smooth out variations in ice thickness across Enceladus.
- **Heat Sources and Their Impact**:** The study identifies some potential heat sources that could keep the ocean in a liquid state, such as hydrothermal vents on the ocean floor and tidal heating from gravitational effects. However, there's still some mystery surrounding which of these sources plays the biggest role in keeping the ocean warm enough to remain liquid. This uncertainty largely stems from our limited understanding of how the ice shell and the core behave under different conditions.
- **Salinity Levels**:** When it comes to the salinity of Enceladus' Ocean, things aren't completely clear. The researchers estimate it could range from 2 to 20 practical salinity units (psu), but they suspect it might often be lower than 10 psu. To keep those intriguing geysers at the south pole active, the salinity needs to be at least around 17 psu. The samples taken from these geysers by the Cassini spacecraft suggest salinity levels between 5 and 20 psu, but the true figures might be even lower due to the way vapor interacts with the environment on the moon.
- **Exciting Astrobiological Potential**:** Understanding these dynamics is crucial for making sense of the geyser samples, which contain fascinating chemicals like CO₂, methane, sodium salts, and even organic materials. This suggests that there's a lively chemical environment beneath the ice that could be conducive to life. Learning how these chemicals work together in the ocean can provide valuable clues about possible biological processes that might be occurring.
- **Modeling Challenges**:** One of the challenges that the researchers face is accurately modeling how heat moves and transforms in this complex system. Current models are having trouble generating enough heat to keep the ice shell as thin as scientists observe it. Moving forward, there's a clear need for more advanced models that can effectively incorporate the different properties of ice under varying conditions.

In a nutshell, this study reveals just how fascinating and complex the interactions between salinity, ocean dynamics, and ice structures are on Enceladus. The research helps paint a clearer picture of the moon's intriguing potential for supporting life, and it sets the stage for more exciting discoveries in the future as we continue to explore what lies beneath its icy surface!

Dynamic Europa Ocean shows transient Taylor columns and convection driven by ice melting and salinity

<https://1drv.ms/b/c/d73ff24855f81e95/ERFgFpiITtBHsNvnf9TujqQB1JOm7ovpCLGHb8eRYq8B6w>

Yosef Ashkenazy and Eli Tziperman explored the ocean beneath Europa's thick ice, aiming to understand how this moon might support extraterrestrial life. Despite its potential, we still know little about the ocean's dynamics. Europa has a deep ocean, roughly 100 kilometers below an ice layer that is several to tens of kilometers thick.

The researchers used a high-resolution General Circulation Model (GCM), known as MITgcm, to study how the ocean behaves. They focused on how heating at the ocean surface and bottom, along with melting and freezing ice, affects salinity. This approach improves upon earlier studies, which did not fully account for salinity and relied on fixed temperature models.

Here are the main findings:

- **Salinity's Importance**: The ocean has weak stratification, primarily influenced by salinity variations. This changes our understanding of how heat and mass move through the ocean.
- **Active Convection**: The ocean displays strong transient convection patterns that create swirling eddies and zonal jets. These movements are key to understanding how heat circulates.
- **Taylor Columns**: The findings show that ocean currents form Taylor columns, which align with Europa's rotation axis. These currents are stable in certain areas and may move toward the equator outside those areas.
- **Heat Transport**: The model indicates that heat transport in the ocean is intense, leading to a nearly uniform thickness of the ice above. We might observe this consistency in future missions to Europa.
- **Better Modeling Techniques**: The researchers used improved methods, allowing the ocean's temperature and vertical changes to be determined by the model itself. This gives a clearer picture of the ocean's temperature dynamics.

Overall, this research highlights the need to consider salinity and non-hydrostatic effects to understand Europa's Ocean better. The findings also suggest that localized convection might help explain the interesting surface patterns seen on the moon, contributing to our understanding of this fascinating world and its potential for life.

Enceladus's oceans may be the right saltiness to sustain life

<https://www.newscientist.com/article/2329739-enceladuss-oceans-may-be-the-right-saltiness-to-sustain-life/>

The way ice covers the surface of Saturn's moon Enceladus suggests that the oceans trapped beneath it may be only a little less salty than Earth's oceans. The finding adds to the possibility that this moon might be able to sustain life.

The surface of Enceladus is encased in clean, bright ice. [Wanying Kang](#) at Massachusetts Institute of Technology and her colleagues wanted to determine what the characteristics of this ice shell indicate about the ocean beneath it.

Samples taken by the [Cassini spacecraft](#) of geyser-like jets of water from Enceladus's surface previously showed that there is some organic matter that could sustain [potential life](#) on the icy moon. Considering the waters under Enceladus's ice was the logical next step for inferring its habitability, says Kang.

The team devised a theoretical model detailing how ocean salinity, ocean currents and ice geometry affect each other on a planet or a moon, then tweaked it to best reproduce the properties of Enceladus's ice.

The researchers found that saltier subsurface oceans correspond to thicker ice on a planet's poles than over its equator and vice versa for less salty water. On Enceladus, the ice over the poles is thinner than the ice over the equator. The specific variation in thickness suggests that the ocean's salinity could be as high as 30 grams of salt in a kilogram of water. For comparison, Earth's oceans have a salinity of 35 grams of salt per kilogram of water.

The researchers also determined details of water circulation under the moon's ice. These currents are related to temperature differences in the water so understanding them is also important for determining habitability, says Kang.

The team found that some heat emanates from the bottom of Enceladus's Ocean, possibly indicating the existence of heat vents in the ocean floor. Kang says that some astrobiologists have previously suggested that, like on Earth, [such hydrothermal vents](#) could be where life is found in the future.

[David Stevens](#) at the University of East Anglia, UK, says that the behaviour of ice and water on other planets is directly related to their habitability. At the same time, salinity is only one factor, he says.

Kang and her team are currently working on applying the new model to Jupiter's moon Europa, whose oceans are thought to have a higher salinity than those on Earth and Enceladus. Ultimately, they want to pin down the details of the oceans of all icy moons and planets observed by space missions as a step towards better determining how habitable they all are.